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Applying NMT-Adapt to Malayalam

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1 Introduction

There has been extensive research into achieving various NLP objectives for low-resource languages. In the case of neural machine translation, the lack of parallel data for low-resource languages is a problem that direly needs to be solved to reduce bias and increase fairness in the digital world. Here, we have specifically chosen to work with a method called NMT-Adapt (Ko et al., 2021). NMT-Adapt is applicable when there exists a pair of linguistically similar languages, one of which is high-resource, say language 'A', and the other being low-resource, say language 'B'. This method adapts a neural machine translation model between a source language, say 'C', and 'A' to one between 'C' and 'B', using monolingual data in 'B'.

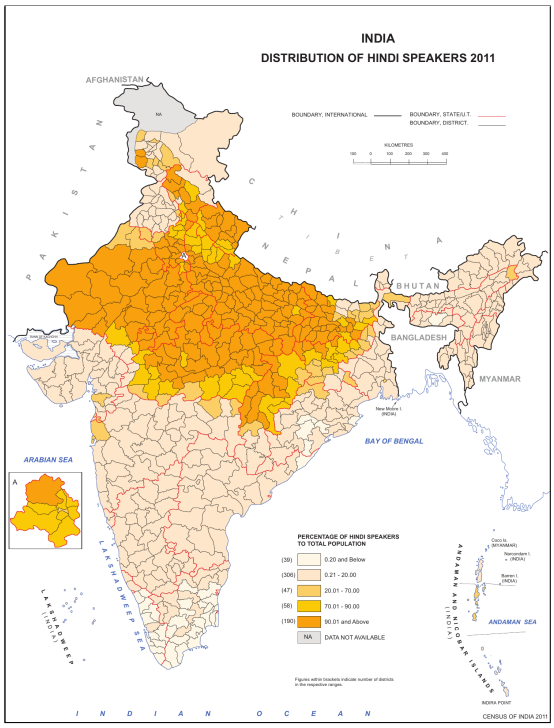
We aim to apply this method for the translation of English to Malayalam with Hindi as the high-resource linguistically similar language. Hindi is an Indo-Aryan language (Office of the Registrar General & Census Commissioner, India (ORGI), 2022a) that is the most widely spoken one in India (see Figure 1a). Malayalam is a Southern Dravidian language spoken mainly in the state of Kerala and the union territories of Lakshadweep and Puducherry in India (see Figure 1b).

1.1 Motivation

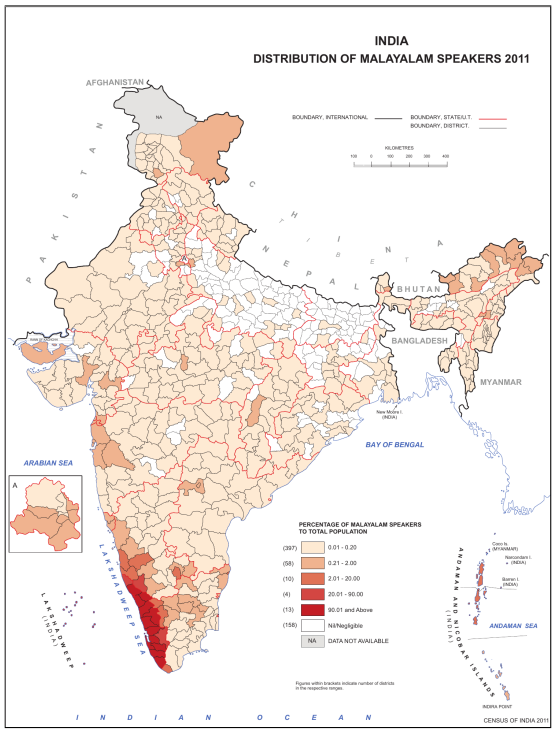
In the original study (Ko et al., 2021), NMT-Adapt was mainly applied to pairs of languages that are either closely related by belonging to the same language family (Spanish and Portuguese for example) or which form dialects or variants of the same language (Hindi and Urdu). However, in linguistically rich and diverse regions such as the Indian subcontinent, there can also be languages that form a *sprachbund*, i.e., they are genetically unrelated but share many features due to geographical proximity and continuous contact throughout history (Emeneau, 1956). Thus, even though languages such as Hindi and Tamil (or Malayalam, which descended from Early Tamil) originate from two different language families, they have borrowed vocabulary from each other and influenced each other for over millennia (Steever, 2017). In addition to loanwords and cultural influences, these languages have also converged to share many linguistic and syntactical features such as following a subject-object-verb order (verb-final), the use of compound verbs, and repetition of the verb stem for emphasis (Subbarao, 2012).

Taking into account this unique type of linguistic similarity, we see an opportunity for methods such as NMT-Adapt to contribute to developing neural machine translation models for low-resource languages belonging to this category. Thus, we have set out to test if NMT-Adapt can be a useful method in the translation between English and a relatively low-resource language such as Malayalam, using its similarity to a high-resource language such as Hindi. To get a better idea about where Hindi and Malayalam are located within the Indian linguistic context, please refer to the language tree[1] depicted in Figure 2 that features a selected set of popular languages under each family. With around 122

[1]The information we used to draw the language tree is from *Ethnologue*, `https://www.ethnologue.com/language/ajp`



(a) Distribution of Hindi Speakers



(b) Distribution of Malayalam Speakers

Figure 1: Distribution of Hindi (Office of the Registrar General & Census Commissioner, India (ORGI), 2022a) and Malayalam speakers (Office of the Registrar General & Census Commissioner, India (ORGI), 2022b) in India taken from the Language Atlas of India 2011

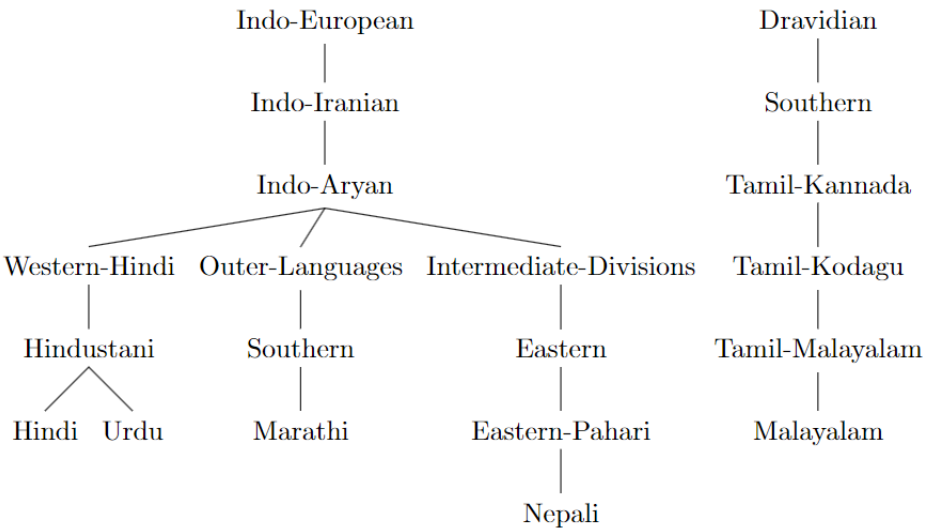


Figure 2: Language tree of Malayalam and languages of Indic language grouping from the experiment in the NMT-Adapt paper(Ko et al., 2021)

major languages, of which 22 languages are listed officially in the constitution, a complete tree of Indian languages would be impractical to include here.

More than half of the population of India (more than 690 million) can speak Hindi whereas Malayalam is spoken by less than 3% of the population (around 36 million), confined mainly to the southern state of Kerala and the island of Lakshadweep. Although relatively less popular in terms of the number of native speakers, Malayalam has a rich literary tradition that is centuries old, making it easier to acquire data in that language.

1.2 Description of Work

In this project[2], we applied the NMT-Adapt method (Ko et al., 2021) to translate English to Malayalam, by taking Malayalam as the low-resource language and Hindi as the high-resource language. We then compared the result of this translation model to baseline models that were trained on English and Malayalam parallel data. Although Malayalam has more parallel data available compared to other languages such as Tulu, we simulate a low-resource scenario by using a small parallel dataset, to test the effectiveness of the NMT-Adapt method in a *Sprachbund* context (Emeneau, 1956). If the NMT-Adapt method is found to be effective, then there is wide scope for applying it to numerous other low-resource languages in the Indian language families which currently do not have any publicly available parallel datasets like the Tulu language.

2 Related Work

In this chapter, we present a summary of the state-of-the-art and relevant background literature.

Zero-shot translation The NMT-Adapt approach is inspired by zero-shot translation. Usually, in zero-shot translation, there is the assumption of a pivot language sharing parallel data with the zero-shot source and target languages. However, the NMT-Adapt model does not use parallel data between the low-resource language and any other language.

Unsupervised translation Unsupervised translation is also closely related to the NMT-Adapt approach. This approach translates between language pairs with no parallel corpus of any kind. In work by Lample et al. (2018), unsupervised translation is performed by training denoising autoencoding and back translation tasks concurrently. Garcia et al. (2021) built a single model for five low-resource languages, which leverages monolingual and auxiliary parallel data from other high-resource language pairs. While these approaches view the parallel data as auxiliary data, to supplement unsupervised translation, the NMT-Adapt approach attempts to use monolingual data in the low-resource language to make the supervised model trained on high-resource language generalize to the low-resource language.

[2]Code and documentation are available at `https://github.com/tianshuailu/NMT-Adapt_ml_IN`

Leveraging High-resource Languages to Improve Low-resource Translation Some work leveraged high-resource language data to improve the translation of similar low-resource languages. Neubig and Hu (2018) proposed methods starting with multilingual models and then continuing training on data related to the low-resource language. Xia et al. (2019) proposed a general framework of data augmentation for low-resource machine translation not only by using target-side monolingual data but also by pivoting through a related high-resource language. However, these previous approaches emphasize translating from the low-resource language to English, while the opposite direction is either unconsidered or shows poor translation performance.

Similar language translation Fan et al. (2021) trained a massively multilingual many-to-many model and demonstrates that high-resource training data improves related low-resource language translation. Wan et al. (2020) investigated how to exploit the commonality and diversity between dialects thus building unsupervised translation models merely having access to monolingual data.

3 Malayalam NMT-Adapt Model

3.1 NMT-Adapt

The NMT-Adapt method (Ko et al., 2021) aims to take advantage of the lexical and syntactic structure similarities between high-resource languages and low-resource languages and train a translation model without using any parallel data. It combines denoising autoencoding, back-translation, and adversarial objectives to utilize monolingual data for low-resource adaptation.

In the paper, they experimented with three groups of languages, which were Iberian languages, Indic languages, and Arabic. In the Indic languages, they treated Hindi as the high-resource language, and Marathi, Nepali, and Urdu as related low-resource languages. In our project, we treated Hindi as the high-resource language and Malayalam as the related low-resource language.

3.2 Apply NMT-Adapt approach to Malayalam

We started with fine-tuning an mBART model on English-Hindi, whose pretraining is described in Liu et al. (2020). One additional step we took compared to the NMT-Adapt paper(Ko et al., 2021) was that we trimmed the mBART model before training with the method mentioned on GitHub[3], since the model is too big to fit on the GPU we used. The method reduced the size of the pretrained model by pruning the word embeddings.

Next, we transliterated the Malayalam script into the Hindi script Devanagari. In the NMT-Adapt paper(Ko et al., 2021), all of the languages they experimented with share the same script as the languages that mBART was trained on, i.e. the dictionary of their

[3] `https://github.com/facebookresearch/fairseq/issues/2120`

	Malayalam	Hindi	English
Back-translation dataset	1 million	-	1 million
Parallel dataset with English	-	800,000	800,000
Denoising autoencoding dataset	1 million	800,000	-
Test dataset	2,000	-	2,000

Table 1: The sizes of the datasets we used. Malayalam denoising autoencoding data contains 1 million parallel noised Malayalam and Malayalam sentences. Hindi denoising autoencoding data contains 800,000 parallel noised Hindi and Hindi sentences.

model already contains all the target language scripts. But in our case, the Malayalam script is not part of the mBART model, hence the dictionary does not include that script. Thus, we transliterated the Malayalam script into Devanagari before feeding the data to the model.

After that, we used the pretrained mBART model to generate English-Malayalam back-translation pairs. At this point, we already pretrained the mBART model on Hindi and English parallel data, then we input the model transliterated Malayalam sentences, set the source language to Hindi, and let the model translate the sentences into English. We used these parallel data to train the model.

Next, we prepared data for denoising autoencoding. This process aims to learn a feature space for the languages such that the encoder and decoder can transform features into sentences. We took Malayalam monolingual data and Hindi sentences from the Hindi-English parallel data. Then we added noise by randomly shuffling the words and masking words. Each word was constrained to shift at most three words from its position, and each word was masked with a uniform probability of 0.1. We input the model with these noised sentences and used the original sentences as target sentences to train the model.

In the next step, we preprocessed the data. The data can be described in four groups as shown in Table 1, which are (1) English Malayalam back translation data, (2) English Hindi parallel data, (3) Malayalam and Hindi denoising autoencoding data, and (4) English Malayalam test data. Note that the Malayalam sentences from the test data are also transliterated. And we also split out 1000 sentences from these training data as validation data, except for the test data. Next, we preprocessed the data with the Fairseq framework (Ott et al., 2019).

In the final step, we trained the NMT-Adapt model in Malayalam, following the instructions from the NMT-Adapt paper's GitHub repository[4]. The training details will be discussed in section 5.3.

[4] `https://github.com/wjko2/NMT-Adapt`

4 Base Model

In this section, we describe the base model for translating English to Malayalam, to compare the performance of the NMT-Adapt method with a straightforward neural machine translation model.

4.1 Model Description

To do a fair comparison, we adopted the same mBART model as a starting point for the base model. Due to limitations in GPU memory, we trimmed the model in the same way as the NMT-Adapt model. As mBART25 is not pre-trained with Malayalam, we again transliterated all the Malayalam data used into Devanagari script. We set a low-resource scenario as a baseline to compare the NMT-Adapt results. Since the definition of "low-resource" is not well-defined, we ran multiple experiments with the training dataset sizes ranging from 1000 to 20000 parallel sentences in English and Malayalam. This range was chosen according to Ko et al. (2021), who applied the NMT-Adapt to languages with parallel dataset sizes from 2000 to 15000 parallel sentences. To observe the improvement in the translation performance with augmenting training data, we trained base models with larger dataset sizes. However, as these larger training dataset sizes are unlikely to occur for low-resource languages, we are mentioning them only for understanding the dependence of the base model architecture on dataset sizes.

4.2 Modelling Pipeline

The base model followed a straightforward modeling pipeline to set a baseline for the results. For each experiment, we transliterated all the Malayalam training, validation, and test data, and trained a sentencepiece model [5] on the training data. We then applied tokenization and byte-pair encoding (BPE) (Sennrich et al., 2016) using the trained sentencepiece model on the entire data. We used Fairseq (Ott et al., 2019) to preprocess, train and evaluate the model, the details of which are described in the following section.

5 Experimental Setup

5.1 Data

5.1.1 NMT-Adapt

All of the parallel data we used are from Samanantar (Ramesh et al., 2022) and the monolingual data from IndicCorp (Kakwani et al., 2020). The data is scraped from

[5] `https://github.com/google/sentencepiece`

web sources, primarily news, magazines, and books. The data is randomly shuffled, untokenized, and deduplicated.

We used parallel Hindi-English data for pretraining the mBART model, 1 million sentences of monolingual Malayalam data for the training of the NMT-Adapt model, and a Malayalam-English parallel dataset of 2,000 sentences as an intermediate test dataset during the training. Furthermore, after completing all the training, we used the test data from Samanantar (Ramesh et al., 2022) as the final test dataset to evaluate the model.

Before using the data, we filtered out the sentences with length more than 200 words or less than three words. Note that the raw data size is much bigger than what we actually used, i.e. the size of the filtered data is the same as we mentioned in the last paragraph.

5.1.2 Base Model

For the base model, we used parallel data in English and Malayalam from Samantar (Ramesh et al., 2022). There are around 5.7 million sentence pairs after removing duplicates in the dataset. After randomly shuffling and filtering out long and short sentences as described for NMT-Adapt, we took the first part of the dataset as training sets for the experiments. For example, for the base model scenario with a training dataset size of 1000 sentences, we took the first 1000 sentences of the cleaned dataset and so on.

For the base model, we considered 22 scenarios. The first 20 scenarios had training dataset sizes ranging from 1000 to 20,000 sentences in increments of 1000. Additionally, we considered two scenarios with training dataset sizes of 500,000 and 1,000,000 sentences.

5.2 Tokenization and Pre-processing

We used Sentencepiece[6] to do the byte-pair-encoding (Sennrich et al., 2016), which does tokenization and subword segmentation. All of the preprocessing, training, and evaluation were done with the Fairseq framework (Ott et al., 2019). For the base model scenarios, we trained separate sentencepiece models for each training dataset, to realistically reflect the data available for tokenization and byte-pair encoding.

5.3 Training Details

5.3.1 NMT-Adapt

We followed most of the training parameter settings from the NMT-Adapt paper(Ko et al., 2021), except for a few small changes due to the memory size of the GPU. First, we set the max tokens per batch to 512 tokens. Next, we set the optimizer to Adafactor (Shazeer and Stern, 2018), which is a stochastic optimization method based on Adam(Kingma and

[6] `https://github.com/google/sentencepiece`

Ba, 2015) that reduces memory usage while retaining the empirical benefits of adaptivity. Besides, we used a memory-efficient version of fp16. At last, we used early stopping and set the patience parameter to 10.

We trained the model on four GPUs and each iteration of training took around one day. Since we trimmed the model, the training took less time than the model in the paper.

We evaluate the performance of the models with SacreBLEU (Post, 2018). BLEU is not a single metric but requires many parameters. These parameters are often not reported or are hard to find, and consequently, BLEU scores between papers cannot be directly compared. So we used SacreBLEU to produce shareable, comparable, and reproducible BLEU scores.

5.3.2 Base Model

As the size of the base model was more manageable compared to the NMT-Adapt model, we kept the maximum tokens per batch unchanged at 1024 and used the Adam optimizer (Kingma and Ba, 2015) with polynomial decay. The models were evaluated with SacreBLEU (Post, 2018), same as the NMT-Adapt model.

6 Evaluation

In this chapter, we will show the performance of the Malayalam NMT-Adapt model and compare it to our baseline model and models from Ko et al. (2021).

6.1 Performance

In this section, we compare our NMT-Adapt model to the models of the Indic languages from the NMT-Adapt paper(Ko et al., 2021) and to a baseline model that is trained on transliterated Malayalam and English parallel data. The baseline model is also trained from a trimmed mBART model, which is the same starting point as our NMT-Adapt model.

As we can see in Table 2, for the first iteration of the training, the NMT-Adapt model has a BLEU score of 2.6. After the second iteration of training, it reaches 4.2. For the baseline model results, the experiments that we conducted simulating low-resource scenarios, i.e., training dataset sizes of up to 20,000 sentences, yielded negligible BLEU scores (the maximum score that we manage to produce was 0.37 for a dataset size of 20,000). Hence, for comparison to the NMT-Adapt results, we are considering the baseline as the model trained on 500,000 sentences of Malayalam and English parallel data. For other languages, they are the baseline models from the paper. We built our own baseline model because the authors of the NMT-Adapt paper (Ko et al., 2021) did not make the code for baseline models publicly available.

Language	Iteration 1	Iteration 2	Baseline
Malayalam	2.6	**4.2**	2.2
Urdu	7.2	**14.6**	5.1
Marathi	16.1	**19.2**	13.1
Nepali	**26.3**	**26.3**	11.9
Egyptian Arabic	**8.0**	6.6	3.3
Levantine Arabic	**5.1**	4.5	1.9

Table 2: BLEU score of our Malayalam NMT-Adapt model compared with some of the other languages from the NMT-Adapt paper (Ko et al., 2021).

English	We are a young Nation.
Target Malayalam	നമ്മുടേതു യുവത്വമാർന്ന ഒരു രാഷ്ട്രമാണ്.
Target Transliterated Malayalam	«unk»मुटेतु युवत्वमार् न्न ओरु राष्ट्रमाण्.
Transliterated Malayalam	हमीद् युवजन«unk»ङळ् क्क् नेरेयुळ्ळताण्.
Malayalam	ഹമീദ് യുവജന«unk»ങള് ക്ക് നേരെയുള്ളതാണ്.

Table 3: Translation example from the Malayalam NMT-Adapt model. Both the second and the third row are target sentences, the third row is the transliteration of the second row and also what we used as model input. Both the fourth and the fifth row are the translation result, the fourth row is the direct output of the model and the fifth row is the transliteration of it.

6.2 Translation results

As shown in Table 3, the first row shows an English source sentence and the second row the target Malayalam sentence. Since the mBART model is not pretrained on any language that uses Malayalam script, we included the transliteration process before feeding the Malayalam data into the model. The third row is the target transliterated Malayalam. The model translates from an English sentence to transliterated Malayalam, the fourth row. The fifth sentence is the transliteration from the fourth row back to Malayalam. The translation does not make much sense, the model just picked up the notion of **young** (യുവത്വം) from the source sentence and translated it into **young people** (യുവജനം).

6.3 Discussion

We believe the main reason why the model does not perform well is that Hindi and Malayalam are not similar enough. As we can see in Table2, the performance of NMT-Adapt in Malayalam is worse than in Urdu, Marathi, and Nepali. We think the reason is that despite the similarities between Hindi and Malayalam, they are still not close enough for the NMT-Adapt method to perform effectively.

As depicted in Figure 2 (Section 1), Hindi belongs to the Indo-European language family, while Malayalam derives from the Dravidian language family. However, as mentioned in the motivation for this project, our main objective while choosing these two languages

was to test the effectiveness of the NMT-Adapt for languages that form a *Sprachbund* despite belonging to different families (Emeneau, 1956).

When we transliterate the Malayalam script into Devanagari and pre-process the data, we noticed there are some unknown tokens in the sentence as shown in Table 3. The reason is that after we transliterate Malayalam to Devanagari, some characters are still prominently used in Malayalam but have become obsolete in modern Hindi, which means the dictionary does not contain these tokens. For example, Hindi used to differentiate between the long 'o' (ओ) and short 'ō' (ओ) sounds in its script, while Malayalam still retains (ഓ and ഒ respectively). However, modern Hindi has simplified this into just one character for the 'o' sound (ओ). Since the meaning of a word changes in Malayalam, if we use only the characters currently used in the Devanagari script, we retain those characters, resulting in unknown tokens after pre-processing.

We have seen that the model performs better in Marathi and Nepali than in Urdu. In the language tree, we see that Urdu is very close to Hindi compared to Marathi and Nepali, but the BLEU score for Urdu is only 14.6, while for Marathi and Nepali it is 19.2 and 26.3 respectively. We hypothesize that the main reason is that the scripts of these languages are different. Hindi, Marathi, and Nepali share the same script Devanagari, while Urdu uses Arabic script. If Urdu was transliterated into the Devanagari script, we might expect the performance of the model to be comparable with Marathi and Nepali or even better.

Furthermore, the models for the Arabic languages do not perform well either, even though Modern Standard Arabic (MSA)[7] shares the same script with Egyptian Arabic and Levantine Arabic. Apart from that, they are also fairly close in the language tree as they both belong to the Afro-Asiatic language family. The NMT-Adapt paper (Ko et al., 2021) did not provide an explanation why the model does not perform as well as for Indic languages. We think the lack of training data might be the reason that it does not perform well. Because they only used 465,000 sentences of MSA data for the training of Arabic languages, while 769,000 sentences of Hindi data for Indic languages.

7 Summary and Conclusions

The NMT-Adapt approach is taking low-resource language machine translation to an extreme, in the sense that it is training the model without using any parallel data either between pivot language (e.g. Hindi) and low-resource language (e.g. Malayalam) or between the source language (e.g. English) and low-resource language (e.g. Malayalam). The approach demonstrates good performance in some of the low-resource languages(like Marathi). However, in the case of Malayalam, we can see that the similarities brought about by belonging to a *Sprachbund* are not enough to warrant good performance.

Not utilizing parallel data is an advantage of the NMT-Adapt approach but also a limitation, which means the approach only performs well when the low-resource language shares the same script with the related high-resource language and they should be close in the

[7]Modern Standard Arabic is the related high-resource language in the Arabic languages experiment in the NMT-Adapt paper (Ko et al., 2021)

language tree. And these are just the necessary conditions. Meeting these requirements does not guarantee desirable performance from the model.

However, we believe that there is still scope and need for methods like NMT-Adapt in a linguistic context like that of the Indian subcontinent. For example, Tulu, a language spoken by over one million people in areas within and bordering the same region as Malayalam, is a true low-resource language that is from the same language family as Malayalam and Kannada. However, we could not find any parallel datasets for Tulu for testing the effectiveness of NMT-Adapt. Considering the rapid growth of publicly available datasets for computational linguistics research, we believe that applying NMT-Adapt or a similar method to Tulu with Malayalam or Kannada as the related high-resource language would be a worthwhile future research topic.

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